

	Work Instruction	Doc. Revision	01
	AXI V810 XXL X-Ray Tube Alignment Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	05/13/2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

ViTrox	Work Instruction	Doc. Revision	01
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- Go to path "C:\Program Files\AXI System\v810\<Rev>\xxlConfig" and open "software.config" file.
Search key "saveallconfirmationimage", set to "True" and save. # <Rev> represent to GUI version.

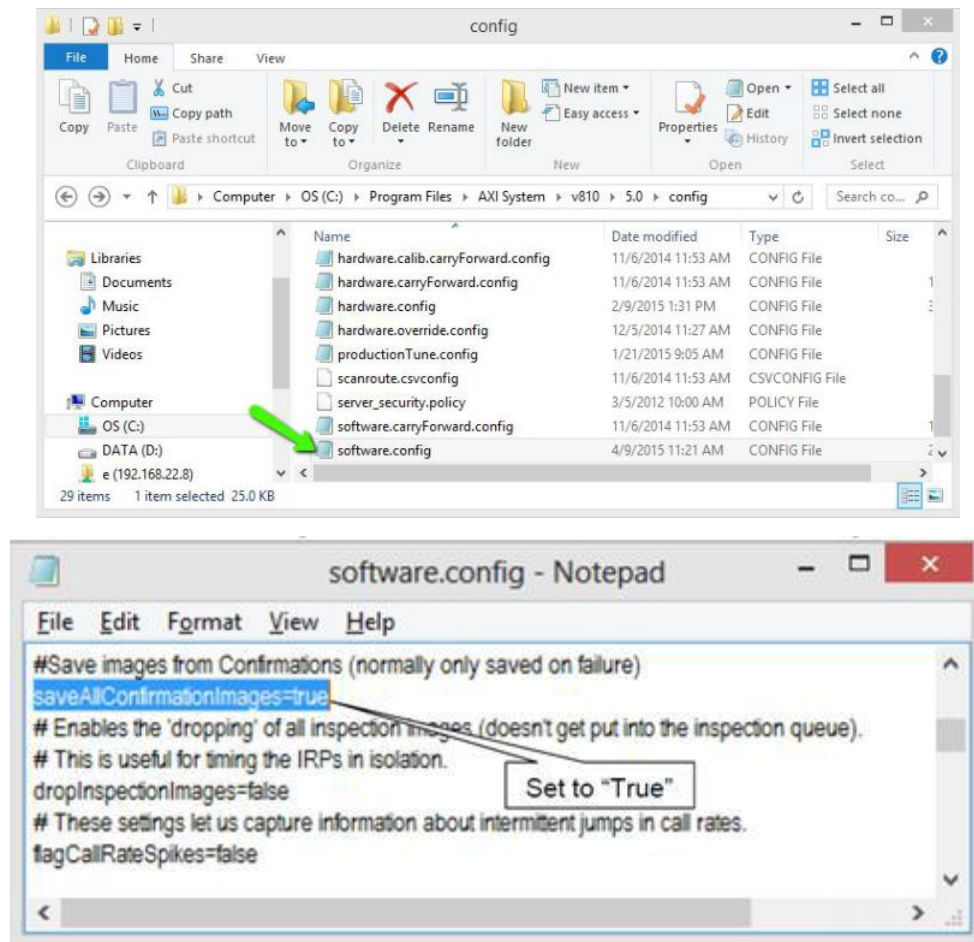


Figure 1

- Go to Connected Processor, click Stop RMS, Upgrade RMS and Restart RMS as Figure 2.
Services → Subsystem Status & Control → Connected Processor → Stop RMS → Upgrade RMS → Restart RMS

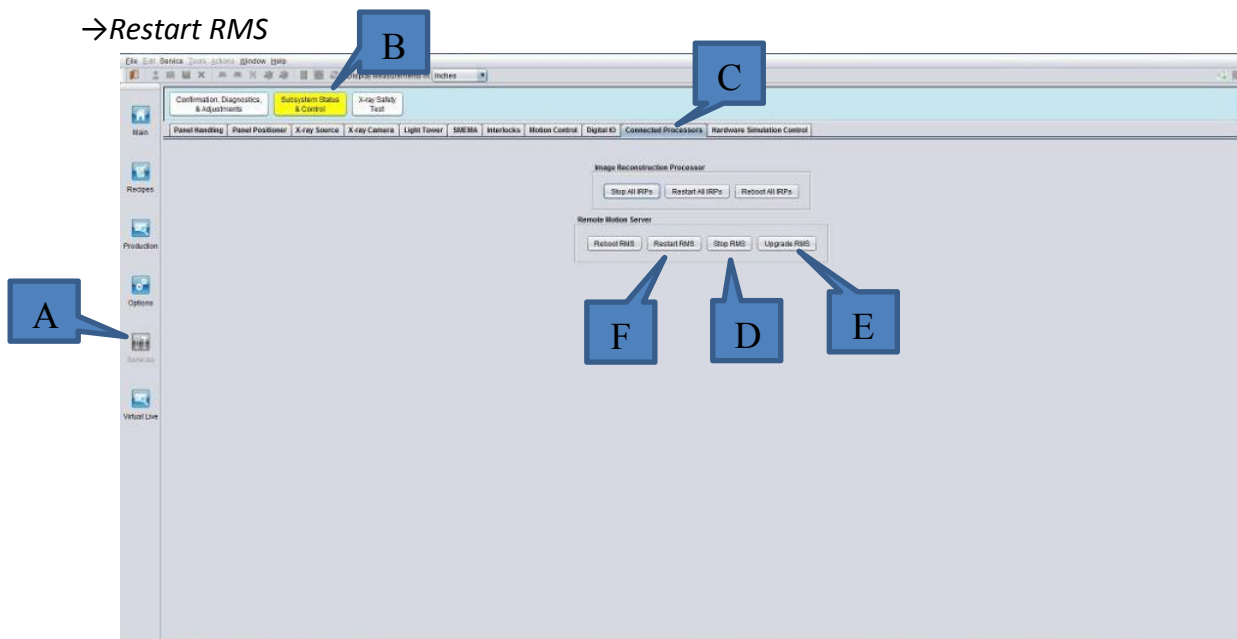


Figure 2

	Work Instruction AXI V810 XXL X-Ray Tube Alignment Procedure	Doc. Revision	01
		Effective Date	2 Oct 2016

3. Quit and restart V810 software. Go to Digital I/O tab and initialize as Figure 3.

Services → Subsystem Status & Control → Digital I/O → Initialize

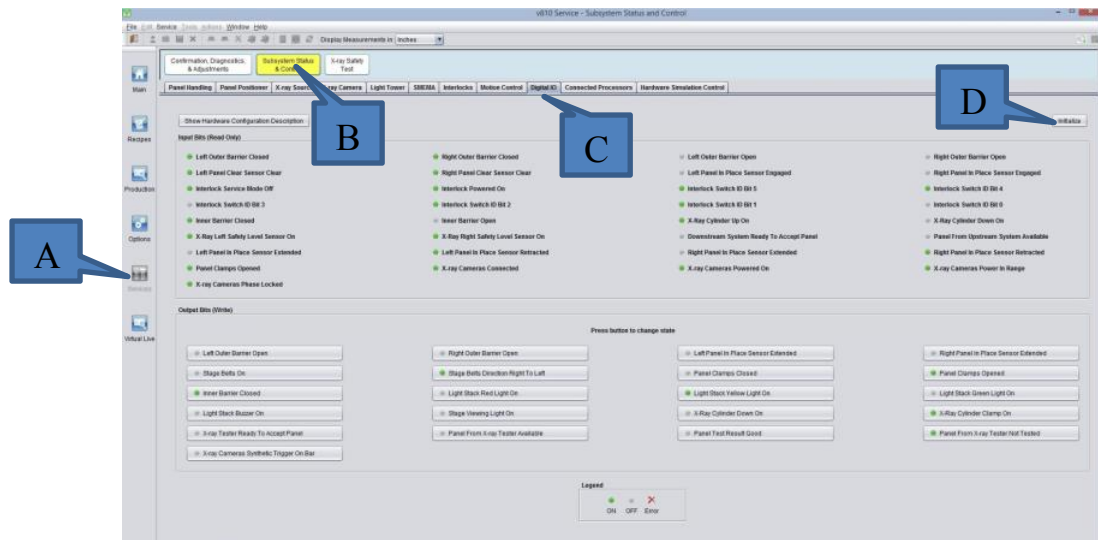


Figure 3

4. Go to Panel Handling tab and initialize as show in Figure 4. Inner Barrier in Open position,

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position

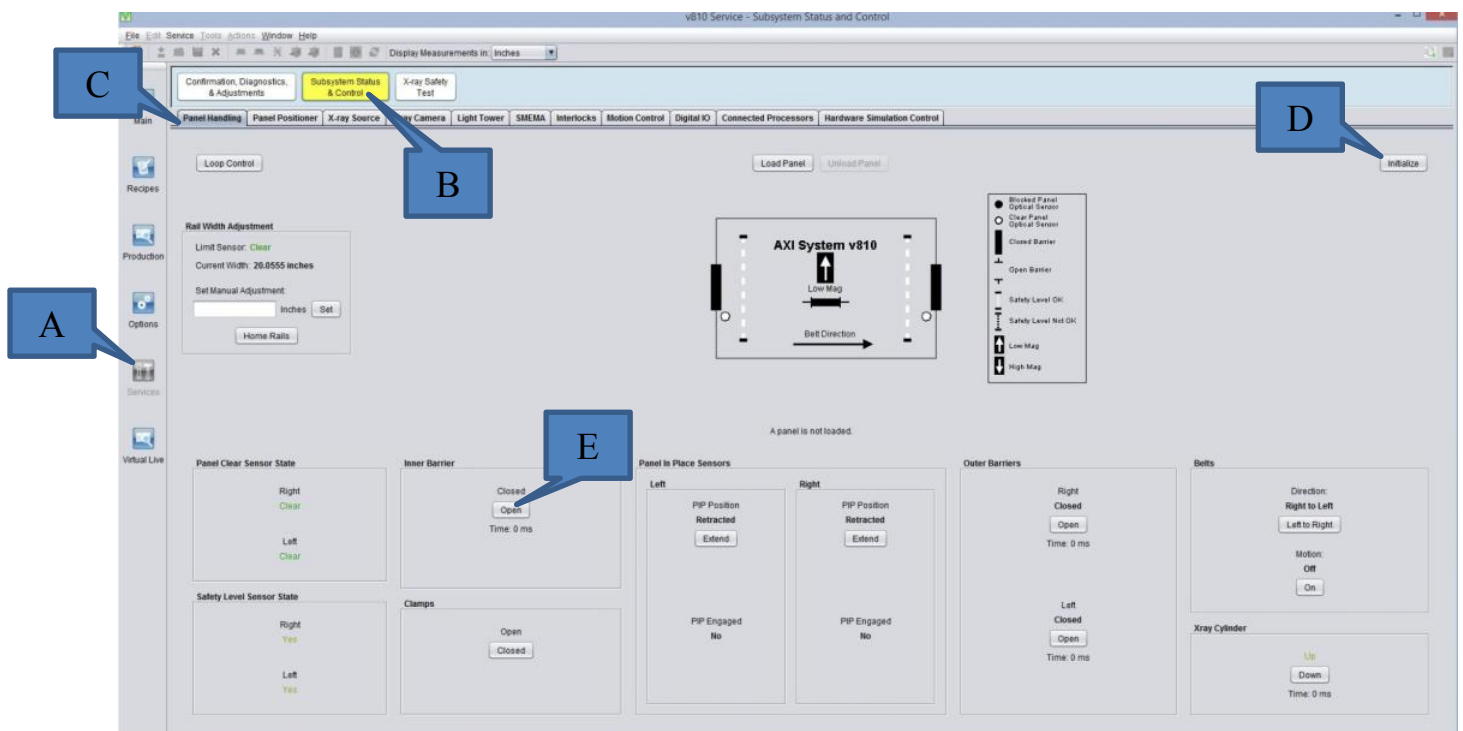


Figure 4

	Work Instruction	Doc. Revision	01
	AXI V810 XXL X-Ray Tube Alignment Procedure	Effective Date	2 Oct 2016

5. Go to Confirmation, Diagnostic and Adjustment tab and perform 1 loop full CD&A task to obtain latest log file as show in Figure 5.

Services → Confirmation, Diagnostic and Adjustment → Scheduled Tasks → Run 1 loop

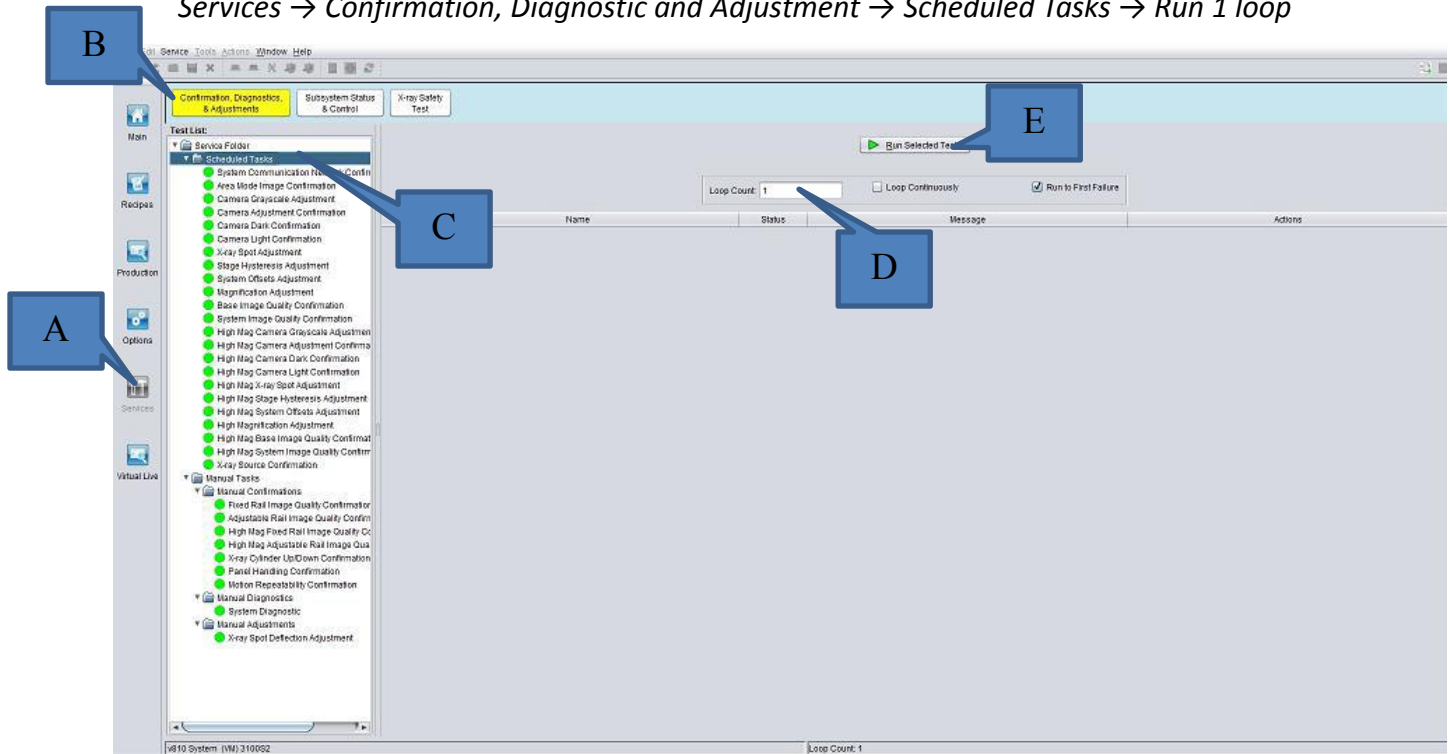


Figure 5

6. Once complete, go to path “C:\Program Files\AXI System\v810\5.0\log\calib\XraySpot” and check for the updated (latest modify date) cam_0.png (X-Ray spot image file). Backup the files as origin reference.

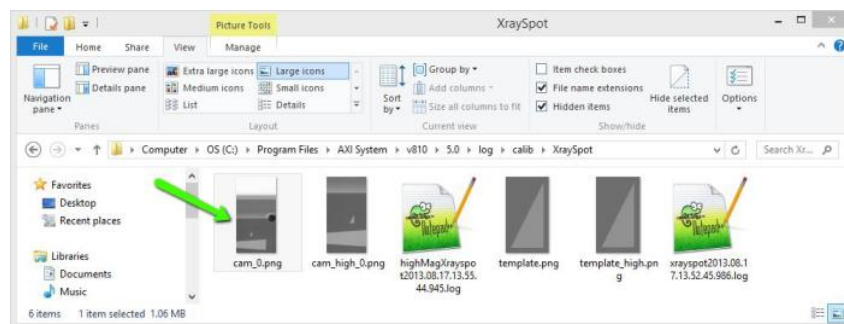


Figure 6

	Work Instruction	Doc. Revision	01
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7. Below is the golden sample for X-Ray spot, compare and adjust the tube position until optimize as image below

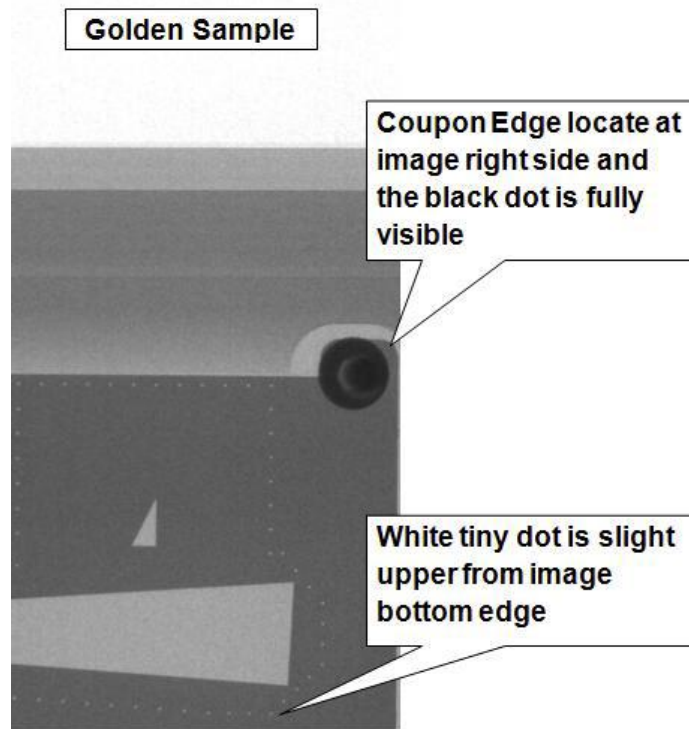


Figure 7

8. **Turn off the X-ray source**,unscrew (Tamper proof screw) as shown and **Open** top access panel

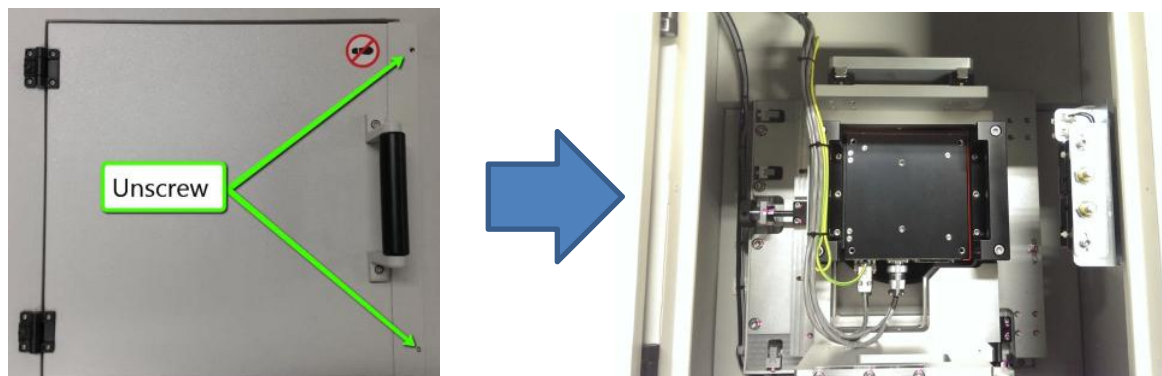


Figure 8

	Work Instruction	Doc. Revision	01
	AXI V810 XXL X-Ray Tube Alignment Procedure	Effective Date	2 Oct 2016

9. Below is the part identify for X-ray tube assembly

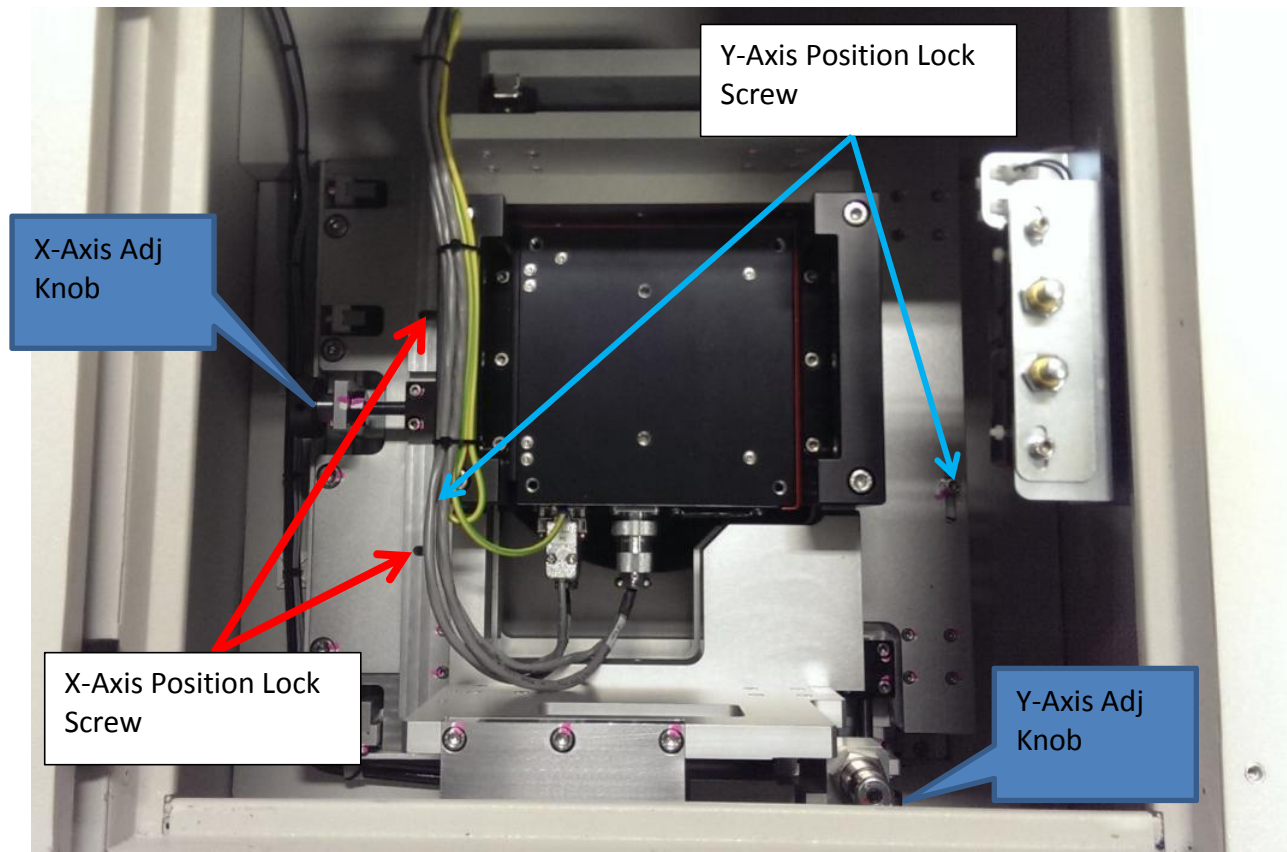


Figure 9

10. In order to move tube position, those lock screws require to be loosen. For the X-axis lock screws, require a special Allen key set (short head and angle) as Figure 10 in order to loosen the screws due to limitation access.



Figure 10

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	AXI V810 XXL X-Ray Tube Alignment Procedure	Effective Date	2 Oct 2016

11. Compare the existing X-ray spot image with the golden sample and move the tube by refer to below guidance

- If image require to move towards **Left direction**, physical **tube position** require move towards **right outer barrier direction**
- If image require to move towards **Upper direction**, physical **tube position** require move towards **front door direction**
- If image require to move towards **Right direction**, physical **tube position** require move towards **left outer barrier direction**
- If image require to move towards **Bottom direction**, physical **tube position** require move towards **rear access door direction**

Sample Case Study

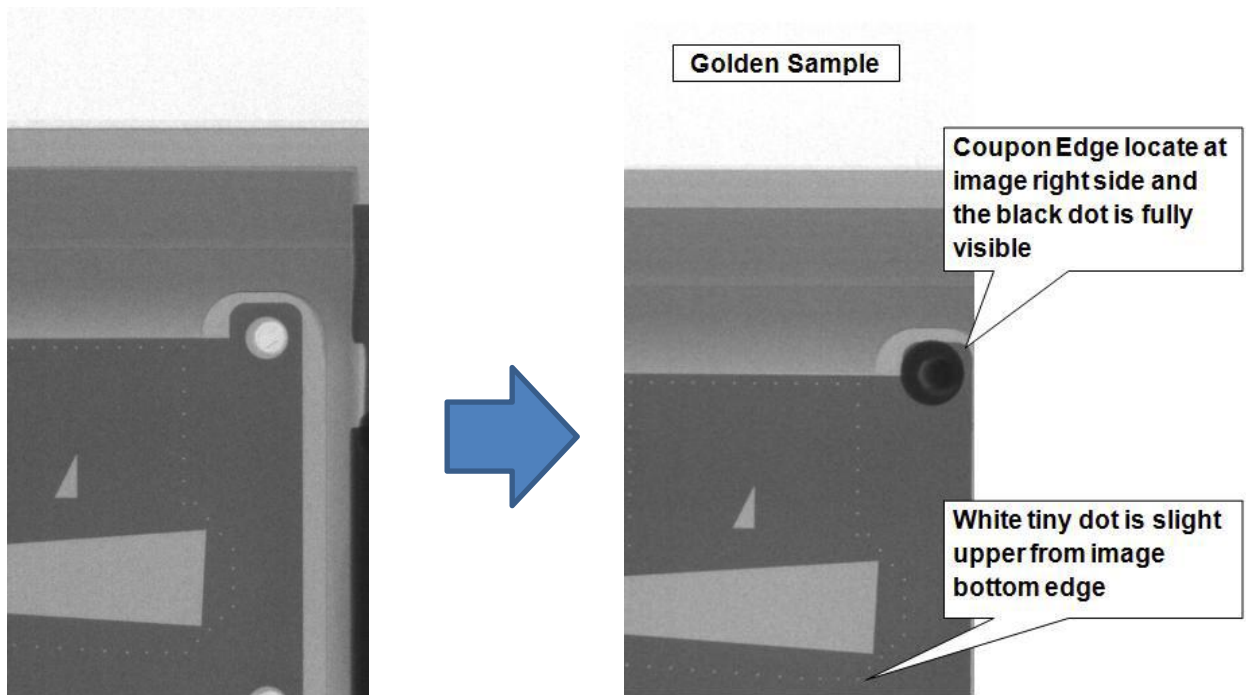


Figure 11

In this case, tube X-Axis require move to machine left side and Y-axis move to machine front side to get the optimize X-Ray spot same as golden sample. Turn the Adj knob to move (# 1 turn of knob = 40 mils (1 mm)) the tube position. Check on the screw track to verify the actual movement direction (X and Y axis)

	Work Instruction	Doc. Revision	01
	AXI V810 XXL X-Ray Tube Alignment Procedure	Effective Date	2 Oct 2016

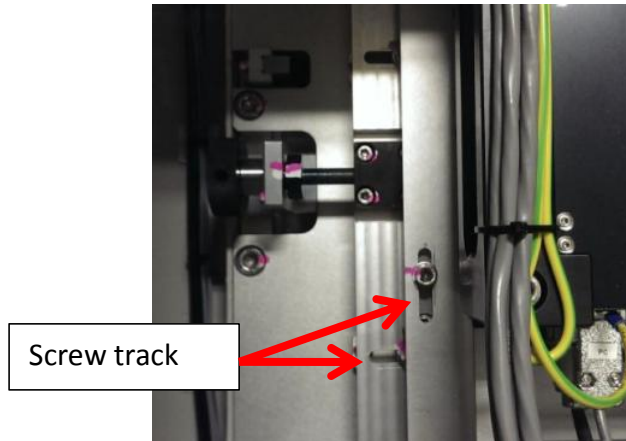


Figure 12

12. Turn the knob 1 turn each time, close the top access door, ON the X-ray source and back to CD&A tab. Select X-Ray Spot Adjustment task and click on “Run test without running Dependent tests”.

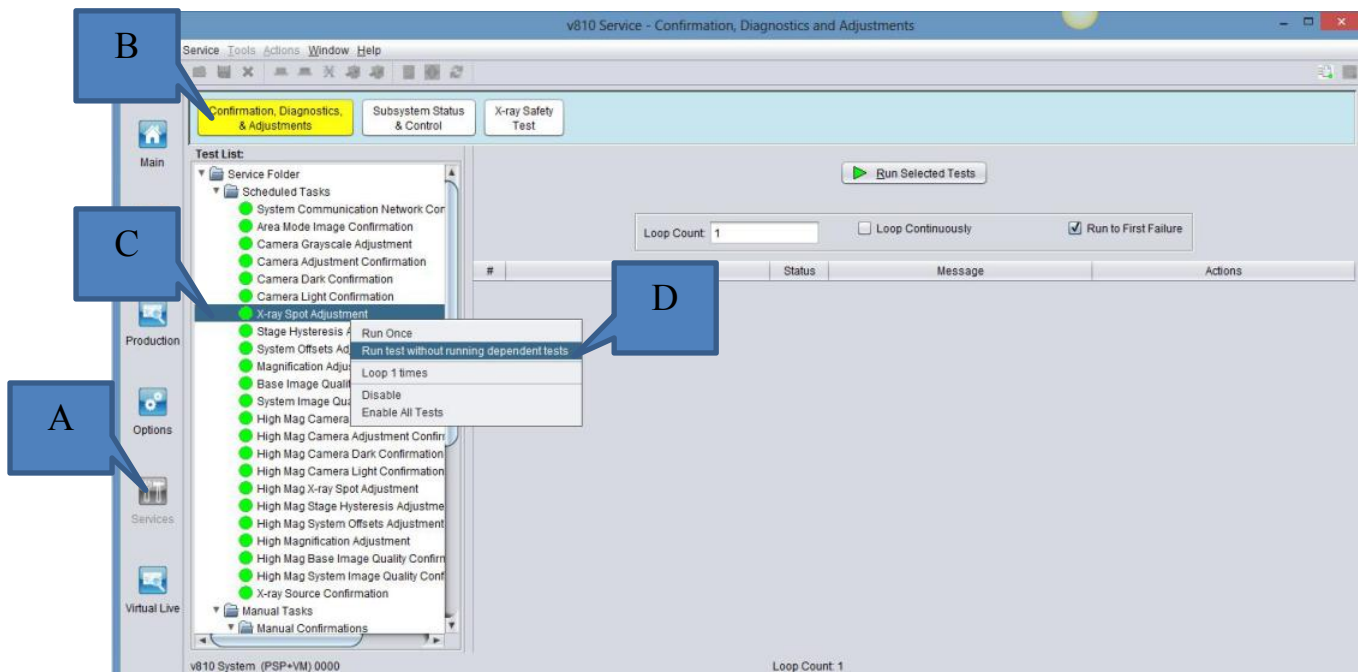


Figure 13

13. Once complete task, check the latest X-ray post image. Repeat step 6 ~ 12 until optimize the X-Ray spot until similar to golden sample below.

	Work Instruction	Doc. Revision	01
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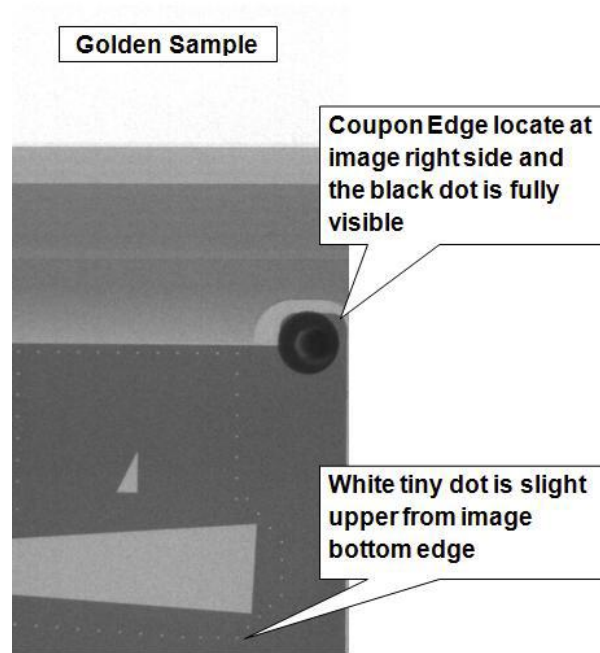


Figure 14

14. Once complete, tighten the locking screws and nuts and close the top access door with screw. Go to Confirmation, Diagnostic and Adjustment tab and perform 5 loop full CD&A task.